# Pocketcreature

An Interactive fiction game on the DTEK -V board

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## Objective

The purpose of this project is to design and implement an interactive fiction game on the DTEK-V board, inspired by the Pokémon games. The player interacts with nine different locations, where each location contains a unique event such as: catching and fighting creatures, interacting with a shop, randomised mysterious events, challenging gyms and defeating the final boss. Captured creatures are used in turnbased fights that follow rock-paper-scissors logic. To complete the game, the player must clear all 9 rooms and defeat a final boss in the last location.

## Project requirements

- Nine different rooms or locations
- One event per location
- Player interaction using I/O devices (switches, LEDs and 7 segment display).
- A non-trivial win condition (defeating the final boss)

## Game specific requirements

In addition to the project requirements, we defined our own design goals:

- Create a coherent and engaging story.
- Include a tutorial sequence, led by Professor Maple.
- Implement a fight and catch system where the player can get either coins through fight or creatures through catch.
- Implement a shop system where coins can be exchanged for creature balls, used to catch creatures.
- Display player and enemy HP on the 7-segment display in battle.
- Through LEDs show the player which room they are in.
- Use polling on the switches to interact with the game when the player is given a choice.

## Solution

The game is implemented in C (and minimally in Assembly) and runs on the DTEK-V board together with its I/O shield. The program is divided into several modules to make the structure clear and maintainable.

The Switches are used for player navigation and menu selection. LEDs indicate the player´s current room and the 7-segment display shows health points during active battle. The story text, dialogues and menus are printed through the JTAG UART terminal. To interface with the DTEK-V hardware, we reused and adapted code from Lab 2 and 3, which provided working drivers for the I/O shield. The existing switch, LED, 7-segment display was modified and integrated into file “io.c” to suit our game.

The file “map.c” defines 9 areas. Each area contains different events such as a shop, gyms, boss, and random encounters with normal creatures throughout. The main gameplay systems are in file “systems.c” : catching, fighting and shopping. Story telling, dialogues, and game UI are handled in file “textstring.c”. Important game data structures, such as the player, items, and creatures, are defined in “objects.c”. And finally, file “main.c” initializes the hardware and game systems, creates the player and map, and starts the gameplay loop.

## Verification

To verify that the system works as intended, both software and hardware testing were performed.

Each software module was tested individually to ensure correct functionality and communication between components. The logic of battles, catching and item usage was validated through repeated playthroughs and debugging output shown in the terminal. During testing, several issues were discovered and resolved:

- Negative HP values sometimes appeared as large positive numbers due to integer underflow which interfered with the alive/dead status in battle.
- The game did not properly terminate after defeating or losing to the final boss, which was fixed by adjusting the game-over logic.
- Switch inputs occasionally triggered multiple times due to signal bouncing, which was solved by adding debounce delays and cleaner polling routines.

We first implemented and tested the program on the computer's terminal to make sure all core game features worked correctly. After gameplay and logic modules were verified, we integrated and tested the input/output parts on the DTEK-V board. This step ensured that all I/O components, such as LEDs, switches, and the seven-segment display, worked as expected and matched the game logic/behavior.

## Contributions

Both team members contribute equally to the implementation of the design, coding and testing of the project.

- **Aleena Amir** focused on implementing the systems in the systems.c file and implementing the input/output handling in io.c, UI and ensuring correct DTEK-V integration.
- **Emilia Lindqvist** designed the story structure, gameloop, events, room logic, and UI.

Both members contributed to coding across multiple files. Both members collaborated on testing, debugging and integrating the full gameplay system.

## Reflections

This project gave further experience in C programming and hardware level development through memory mapped I/O on the DTEK-V board. It taught us how to structure a larger program into clearer modules and how to combine software logic with physical hardware feedback. One of the main difficulties was integrating our independently created files into one program. Through modular programming we refactored the code into smaller well defined modules, which made the project easier to stitch together, maintain and test. Another technical challenge involved input stability and timing on the DTEK-V board. Overall, creating Pocketcreature allowed us to combine creativity, game design, and embedded systems in a single project using our coding and hardware knowledge in one program.